

Lecture 14 Feb 18th

Hydrogen atom:

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{\alpha}{r};$$

Natural units $c = 1; \hbar = 1$

$$\left\{ \begin{array}{l} a_0 = \frac{1}{\alpha m_e}; \quad E_n^{(10)} = -\frac{\alpha^2 m_e}{2n^2} = -\frac{\alpha}{2} \frac{1}{a_0 n^2} \\ \alpha = \frac{1}{137}; \quad m_e = 511 \text{ KeV} \approx 0.5 \text{ MeV} \quad \frac{\hat{p}^2}{2m n^2} \\ \frac{1}{a_0} \approx \text{few KeV} \rightarrow \text{atomic physics} \end{array} \right.$$

$$\Psi_{nlm} = R_{nl}(r) Y_{lm}(\theta, \phi)$$

$$\rho = \frac{r}{na_0}$$

$$= C \times e^{-\rho/2} \rho^l \underbrace{P_{n-l-1}^{2l+1}(\rho)}_{\text{"modes"}} Y_l^m(\theta, \phi)$$

"modes" $\rightarrow$ Polynomial of deg. $N = n - (l+1)$

	s ($\ell = 0$)	p ($\ell = 1$)				d ($\ell = 2$)					f ($\ell = 3$)							
	$m = 0$	$m = 0$	$m = \pm 1$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = \pm 3$			
	s	p _z	p _x	p _y	d _{z²}	d _{xz}	d _{yz}	d _{xy}	d _{x²-y²}	f _{z³}	f _{xz²}	f _{yz²}	f _{xyz}	f _{z(x²-y²)}	f _{x(x²-3y²)}	f _{y(3x²-y²)}		
$n = 1$																		
$n = 2$																		
$n = 3$																		
$n = 4$																		
$n = 5$										...	...	...	...	...	...	...	...	
$n = 6$					...†	...†	...†	...†	...†	...*	...*	...*	...*	...*	...*	...*	...*	
$n = 7$		...†	...†	...†	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	...*	

Nodes

$$N = n - l - 1$$

$$l = 0, \dots, n-1 \rightarrow n$$

$$m = -l, \dots, l \rightarrow \underline{2l+1}$$

Degenerate states:

$$\sum_{l=0}^{n-1} (2l+1) = n^2$$

Represented by $|n l m\rangle$

$$\langle \hat{r} | 100 \rangle = \frac{e^{-r/a_0}}{\sqrt{\pi a_0^3}}$$

BTW: $\langle \hat{r} | \hat{p} | 100 \rangle = -i \nabla \Psi_{100}(r) = -i \partial_r \Psi_{100}(r)$
 $= \frac{i}{a_0} \Psi_{100}(r)$

Typically $\rho \sim \frac{1}{a_0} \sim \alpha m_e \hat{=} m_e v$

$\Rightarrow v \approx \alpha \rightarrow$ non-rel

Does $v \rightarrow 1$ when $Z\alpha \rightarrow 1$?

Missing quantum number: m_s e spin!

(s, m_s) : $s = \frac{1}{2}$; $m_s = \pm \frac{1}{2}$

See H:

$$\hat{L}^2 |n l m\rangle = l(l+1) |n l m\rangle$$

$$\hat{L}_z |n l m\rangle = m |n l m\rangle \quad \text{3/4}$$

Enhance to $\hat{S}^2 |n l m_l m_s\rangle = \frac{1}{2}(\frac{1}{2}+1) |n l m_l m_s\rangle$

Uncoupled basis $\rightarrow \hat{L} \cdot \hat{S}$? ^{Spin} _{Orbit.}

Now, the problem:

Suppose we have a perturbation that depends on the total angular momentum:

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

$$\hat{\mathbf{J}}^2 |j m_j\rangle = j(j+1) |j m_j\rangle$$

$$\hat{J}_z |j m_j\rangle = m_j |j m_j\rangle,$$

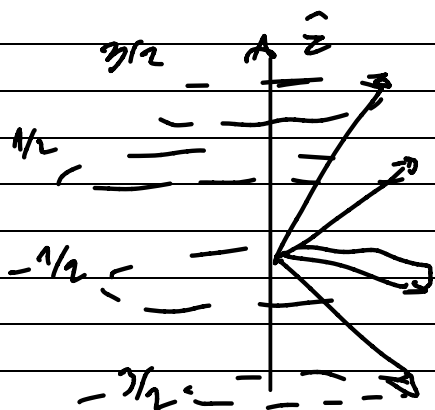
This is a better basis. But

how is it related to $|n l m_l m_s\rangle$?

Note that $\hat{J}_z = \hat{L}_z + \hat{S}_z$

$$[\hat{\mathbf{J}}^2, \hat{J}_z] = 0; \quad [\hat{J}_i, \hat{J}_j] = i \epsilon_{ijk} \hat{J}_k$$

$$\epsilon_{xyz} = +1$$



$$\left\{ \begin{array}{l} \mathcal{H} \approx \mathcal{H}_L \otimes \mathcal{H}_S \\ \dim(\mathcal{H}_L) = 2l+1 \\ \dim(\mathcal{H}_S) = 2 \end{array} \right.$$

$$|n l m_l "s" m_s\rangle = |n l m_l m_s\rangle$$

$$\hat{L}^2, \hat{L}_z, \hat{S}^2, \hat{S}_z.$$

I want to find the basis that diagonalizes $\hat{J}$.

$$\hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_z$$

$$|n l "s" j m_j\rangle = |n l j m_j\rangle$$

⚠ Addition of angular momentum: SU(2)

$$\begin{cases} (2l+1) \otimes 2 = (2l+2) \oplus (2l) & \text{dimension} \\ \text{or} \\ l \otimes \frac{1}{2} = (l+\frac{1}{2}) \oplus (l-\frac{1}{2}) & \text{spin label} \end{cases}$$

BTW:

$$\bullet l=0 \begin{cases} 1 \otimes 2 = 2 \oplus 0 = 2 \\ 0 \otimes \frac{1}{2} = \frac{1}{2} \end{cases}$$

$S_{1/2}$

$$\bullet l=1 \begin{cases} \text{Reducible} & \text{Irreducible} \\ 3 \otimes 2 = 4 \oplus 2 \\ 1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \end{cases}$$

$P_{3/2} \oplus P_{1/2}$

$$\bullet l=2 \begin{cases} 5 \otimes 2 = 6 \oplus 4 \\ 2 \otimes \frac{1}{2} = \frac{5}{2} \oplus \frac{3}{2} \end{cases}$$

$D_{5/2} \oplus D_{3/2}$

$\vdots$

$\vdots$

	S $l=0$	P $l=1$	D $l=2$
$n=4$	<u>4 $S_{1/2}$</u>	<u>4 $P_{3/2}$</u> 4 $P_{1/2}$	<u>4 $D_{3/2}$</u> 4 $D_{5/2}$
$n=3$	<u>3 $S_{1/2}$</u>	<u>3 $P_{3/2}$</u> 3 $P_{1/2}$	<u>3 $D_{3/2}$</u> 3 $D_{5/2}$
$n=2$	<u>2 $S_{1/2}$</u>	<u>2 $P_{3/2}$</u> 2 $P_{1/2}$	
$n=1$	<u>1 $S_{1/2}$</u>		

always $\frac{1}{2}$

because

$$1 \otimes 2 = 2$$

always $\frac{3}{2} + \frac{1}{2}$

because

$$3 \otimes 2 = 4 \oplus 2$$

change of basis:

$$|n \ell j m_j\rangle = \sum_{m_\ell m_s} |n \ell m_\ell m_s\rangle \underbrace{\langle n \ell m_\ell m_s | j m_j \rangle}_{C_{\ell m_\ell, s m_s}^{j m_j}} \quad \delta m_j; m_\ell + m_s$$

Note that $m_j = m_s + m_\ell$

Proof:

$$\hat{J}_z | \ell m_\ell m_s \rangle = (S_z + L_z) | \ell m_\ell m_s \rangle = (m_\ell + m_s) | \ell m_\ell m_s \rangle$$

$$\hat{J}_z | \ell j m_j \rangle = m_j | \ell j m_j \rangle$$

$$\Rightarrow \boxed{\langle \ell m_\ell m_s | \hat{J}_z | \ell j m_j \rangle = m_j C_{\ell m_\ell, s m_s}^{j m_j}} \\ = (m_\ell + m_s) C_{\ell m_\ell, s m_s}^{j m_j}$$

$$\Rightarrow m_j = m_\ell + m_s.$$

Perturbation

Spin - Orbit Coupling

$$H = H_0^{\text{FS}} + \frac{\alpha^2}{2} j(r) \hat{L} \cdot \hat{S}$$

$$= \frac{1}{2m} \left[\hat{p}_r^2 + \frac{\hat{L}^2}{r^2} \right] + V(r) + \frac{\alpha^2}{2} j(r) \hat{L} \cdot \hat{S}$$

$$\hat{p}^2 = \hat{p}_r^2 + \frac{\hat{L}^2}{r^2}$$

$$\text{recall: } \left[-\hbar^2 \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{r^2} \right] \psi_{nlm}$$

$$\hat{J}^2 = \hat{L}^2 + \hat{S}^2 + 2\hat{L} \cdot \hat{S} \Rightarrow \hat{L} \cdot \hat{S} = \frac{\hat{J}^2 - \hat{L}^2 - \hat{S}^2}{2}$$

$$\begin{aligned} \hat{L} \cdot \hat{S} |l j m_j\rangle &= \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2) |l j m_j\rangle \\ &= \frac{1}{2} \left[j(j+1) - l(l+1) - \overset{3/4}{s(s+1)} \right] |l j m_j\rangle \end{aligned}$$

BTW $\hat{L}^2, \hat{S}^2, \hat{J}^2$ all commute w/ $(\hat{L} \cdot \hat{S})$ and $\hat{J}^2$

$$[\hat{L}^2, L_i] = 0; [\hat{S}^2, S_i] = 0 \Rightarrow$$

$$\text{But } [\hat{L} \cdot \hat{S}, L_z] \neq 0$$

$$[\hat{L} \cdot \hat{S}, S_z] \neq 0$$

→ M_s and M_L are not good Quantum numbers.

$$\text{But } [\hat{L} \cdot \hat{S}, J^2] = [L \cdot S, L_z + S_z]$$

$$= \frac{1}{2} ([J^2, J^2] - [L^2, (L_z + S_z)] - [\hat{S}^2, (L_z + S_z)]) = 0$$

⇒ Already Known

$$E_n^{\text{so}} = \langle n l j m_j | \frac{\alpha^2}{2} \delta(r) \hat{L} \cdot \hat{S} | n l j m_j \rangle$$

$$= \frac{\alpha^2}{2} \langle \delta(r) \rangle \left(\frac{j(j+1) - l(l+1) - \frac{3}{4}}{2} \right)$$

$$\vec{\mu}_L = -\frac{e \hbar}{2m_e} \vec{L}$$

$$\vec{\mu}_S = -\frac{g_e e \hbar}{2m_e} \vec{S}$$

$$V = -\vec{\mu} \cdot \vec{B}$$

$$\hat{V}_L \longrightarrow E_{m_L}^{(1)} = \mu_B B m_L$$

$$\hat{V}_S \longrightarrow E_{m_S}^{(1)} = \mu_B B m_S$$

$\vec{B}_L$ induced by nucleus orbiting electron:

$$\hat{V}_{LS} = -\vec{\mu}_S \cdot \vec{B}_L$$

what is B_L ? $B_L = -\vec{r} \times \vec{E}$

$$= -\frac{1}{e m_e} \left(\frac{1}{r} \frac{dV}{dr} \right) \hat{L}$$

$$\frac{dV}{dr} = + \frac{\alpha Z}{r^2} \Rightarrow B_L = \frac{\alpha Z}{e r^3 m_e}$$

$$\langle g(r) \rangle = \int_0^\infty R_{nl}^2(r) \frac{1}{r} \frac{dV}{dr} r^2 dr$$

Lecture 15: Feb 20th

Correcting the Hydrogen atom:

1) Fine Structure: Rel corr.

$$E_n^{(1)}(\text{Rel}) = -\frac{(E_n^{(0)})^2}{2Mc^2} \left[\frac{4n-3}{l+1/2} \right] = E_n^{(0)} \alpha^2 \left[\frac{1-3}{n(l+1/2)4m^2} \right]$$

$$\left(E_n^{(0)} = -\frac{\alpha^2 m_e}{2n^2} = -\frac{\alpha}{2a_0 n^2} \right);$$

	$l=0$	$l=1$
$n=2$	<u>$2S_{1/2}$</u>	<u>$2P_{3/2}$</u> $2P_{1/2}$

$$3 \otimes 2 = 4 \oplus 2$$
$$1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$$

For example:

$$E_1^{(1)}(\text{Rel}) = -\frac{E_1^2}{2Mc^2} (8-3) = -\frac{E_1^2}{2Mc^2}$$

1b) Fine Structure 2: spin orbit coupling.

$$V_{so} = -\vec{\mu}_s \cdot \vec{B}_{int} + \frac{1}{2} \rightarrow \text{THOMAS precession}$$

$$\vec{B}_{int} = -\vec{v} \times \vec{E} = + \frac{\vec{E} \times \vec{p}}{m}$$

$$\text{with } \vec{F} = -e\vec{E} = -\frac{dV(r)}{dr} \frac{\vec{r}}{r}$$

$$\Rightarrow \vec{B}_{int} = -\frac{1}{r m} \frac{dV(r)}{dr} \underbrace{\vec{r} \times \vec{p}}_{\vec{L}}$$

$$\Rightarrow \vec{\mu}_s = \frac{g_e e}{2m} \vec{S} ; V(r) = \frac{-\alpha}{r}$$

$$\Rightarrow V_{so} = \frac{g_e}{4m^2} \times \frac{1}{r} \times \frac{dV}{dr} \vec{S} \cdot \vec{L}$$

$$= + \frac{g_e \alpha}{4m^2 r^3} \vec{S} \cdot \vec{L} = + \frac{g_e E_0 \alpha}{2} \frac{Q_0}{r^3} \vec{S} \cdot \vec{L}$$

$$m = \frac{1}{\alpha a_0}$$

$$E_0 = -\frac{m \alpha^2}{2}$$

$$V_{so} = \alpha \frac{g_e}{2} f(r) \vec{L} \cdot \vec{S} \Rightarrow$$

$$f(r) = \left(\frac{g_e E_0 Q_0}{r^3} \right)$$

Now we are ready:

$$E_{mlj}^{(1)}(SO) = \langle n l j m_j | V_{SO} | n l j m_j \rangle$$

Recall $L \cdot S = \frac{1}{2} (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2)$

$\frac{1}{2} l ; j = l + \frac{1}{2}$
 $-\frac{(l+1)}{2} ; j = l - \frac{1}{2}$

$$\Rightarrow \mathbf{L} \cdot \mathbf{S} | n l j m_j \rangle = \frac{1}{2} (j(j+1) - l(l+1) - \frac{3}{4}) | n l j m_j \rangle$$

$$\Rightarrow E_{mlj}^{(1)}(SO) = \frac{\alpha^2}{2} g_e \underbrace{\langle L S \rangle}_{0, \neq} \underbrace{\left\langle \frac{1}{r} \right\rangle}_r$$

$$\left\langle \frac{1}{r} \right\rangle = \int_0^\infty dr \frac{R_{nl}^2(r)}{r} = \frac{1}{l(l+\frac{1}{2})(l+1)n^3 a_0^3} \quad (l > 0)$$

for $l=0$, $E_{m0\frac{1}{2}} = 0$ since no orbit for spin to couple to.

$$\Rightarrow E_{mlj}^{(1)}(SO) = \frac{\alpha^2}{2} g_e \left(\frac{E_1^{(0)}}{m^2} \right) \left[\frac{j(j+1) - l(l+1) - \frac{3}{4}}{l(l+\frac{1}{2})(l+1)n} \right] \quad l > 0$$

so far, FS corrections:

$$E_{mlj} = E_M^{(0)} \left[1 + \alpha^2 \left(\frac{1}{n(l+\frac{1}{2})} - \frac{3}{4m^2} + \frac{j(j+1) - l(l+1) - \frac{3}{4}}{2l(l+\frac{1}{2})(l+1)n} \right) \right]$$

$$= E_M^{(0)} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4} \right) \right] \quad \text{if } l > 0 \quad \text{Next Darwin.}$$

$2P_{1/2} / 2P_{3/2}$

 $\frac{3}{2}$

 $\frac{1}{2}$

15 1/2

$$V_D = \frac{1}{8m^2} \nabla^2 V(R) = \frac{\pi \alpha}{2m^2} \delta(\vec{r})$$

Easy ...

$$E^{(1)} = \langle n l m | \hat{V}_D | n l m \rangle$$

$$= \frac{\alpha \pi}{2m^2} |\psi_{nm}(0)|^2 = 0 \quad l=0; m=0$$

Recall $\psi_n(\vec{r}) = C \times \rho^l e^{-\rho/2} L_{n-l-1}^{2l+1} \dots$

$$\Rightarrow E_{n=0}^{(1)} = \frac{\alpha \pi | \psi(0) |^2}{2m^2} \xrightarrow{n=0} \frac{1}{\pi n^3 a_0^3} = \frac{\alpha}{2m^2 n^3 a_0^3}$$

$$= \frac{\alpha^4 \mathcal{M}}{2n^3} = -E_n^{(0)} \frac{\alpha^2}{n}$$

Same Order in α as S.O. and rel.

Hyperfine :

$$\vec{\mu}_p = \frac{g_p e}{2m_p} \vec{S}_p \quad \mu_e = -\frac{g_e e}{2m_e} \vec{S}_e$$

$$g_p = 5.58 \rightarrow g\text{-factor} \leftarrow g_e \approx 2$$

$$\vec{B} = \left(\frac{3 (\vec{\mu} \cdot \hat{r}) \hat{r} - \vec{\mu}}{4\pi r^3} \right) + \frac{2}{3} \vec{\mu} \delta^{(3)}(\vec{r})$$

$$H = \frac{g_p \alpha}{m_p m_e} \left[\frac{3 \vec{S}_p \hat{r} \vec{S}_e \hat{r} - \vec{S}_p \vec{S}_e}{r^3} \right]$$

$$+ \frac{g_p \alpha}{3m_p m_e} \vec{S}_p \cdot \vec{S}_e \delta^{(3)}(\vec{r})$$

$$E^{(1)}(\text{HFS}) = \frac{g_p 4\pi \alpha}{3m_p m_e} \underbrace{\langle S_p \cdot S_e \rangle}_{?} |\psi(0)|^2$$

$$\vec{S} = \vec{S}_e + \vec{S}_p \Rightarrow S_p \cdot S_e = \frac{S^2 - S_p^2 - S_e^2}{2}$$

$$\vec{S}_p^2 |m_l m_s m_p\rangle = \frac{3}{4} |\dots\rangle;$$

$$S_p \cdot S_e |n l m_s\rangle = \frac{1}{2} \left(S^2 - 2 \times \frac{3}{4} \right)$$

$$= \frac{S^2 - 3}{2}$$

$$|m_l m_s\rangle \rightarrow |S m_s\rangle$$

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$$

$$2 \otimes 2 = 3 \oplus 1$$

$\swarrow$ Singlet
 $\searrow$ Triplet

$$\hat{S}^2 |S m_s\rangle = S(S+1) |S m_s\rangle$$

$$m_s = m_l + m_p = 1 \text{ or } 0$$

$$S = 0 \text{ or } 1$$

$$\begin{aligned}
 \Rightarrow S_p \cdot S_e |n l m_s, S m_s\rangle &= +\frac{1}{4} \quad \text{if } S=1 \\
 &= -\frac{3}{4} \quad \text{if } S=0
 \end{aligned}$$

$$\Rightarrow E^{(1)}(\text{HFS}) = \frac{g_p 4 \alpha}{3 m_p m_e} \frac{\langle S_p S_e \rangle}{n^3 a_0^3}$$

$S=1$
 $S=0$

$$= \frac{4 g_p}{3 m_p} \cdot \frac{\alpha^4 m_e^2}{n^2} = \frac{4 g_p}{3} \frac{m_e}{m_p} \frac{E_n^{(0)}}{n} \frac{\alpha^4}{n} \langle S_p S_e \rangle$$

$$F = J + I; (F - J)^2 = I^2 \\ \Rightarrow \frac{F^2 + J^2 - I^2}{2} = J \cdot F$$

$$\begin{aligned} A \cdot J &= \frac{(A \cdot J)}{j(j+1)} \frac{(I \cdot F)}{f(f+1)} (J \cdot F) \\ &= \frac{L^2}{\pi^3} \left(\frac{F^2 - J^2 - I^2}{2} \right) \left(\frac{F^2 + J^2 - I^2}{2} \right) \left[\frac{1}{j(j+1)f(f+1)} \right] \\ &= \frac{L^2}{4\pi^3} \left((F^2 - I^2)^2 - J^4 \right) \frac{1}{(\dots)} \end{aligned}$$

$$\cdot \text{Try } I \cdot J = \frac{F^2 - J^2 - I^2}{2}$$

$|n, l, j, m_f\rangle$ states

$$\langle A \cdot I \rangle = \frac{l(l+1)(f(f+1) - j(j+1) - i(i+1))}{2\pi^3 \times j(j+1)} \left\langle \frac{1}{\pi^3} \right\rangle$$

$$|n, l, j, m_f\rangle = |1, 0, \frac{1}{2}, m_f\rangle //$$